



MODELING AND CHARACTERIZATION OF LARGE-SCALE WI-FI TRAFFIC IN PUBLIC HOT-SPOTS

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OUTLINE



- ☐ **Goals**
- ☐ **An Overview of Data**
- ☐ **Arrival Count Modeling**
- ☐ **Connection Duration Modeling**
- ☐ **Simultaneous Users Modeling**
- ☐ **Conclusions**

GOALS



❑ **Study and analyze** Wi-Fi traces collected at different venues by AT&T in March 2010 in NYC and SF

- Coffee shops, fast food chains, book stores, hotels, enterprises, etc
- Data contains:
 - Connection login/logout times
 - Bytes uploaded/downloaded
 - Venue size (small, medium, large), zip codes, etc

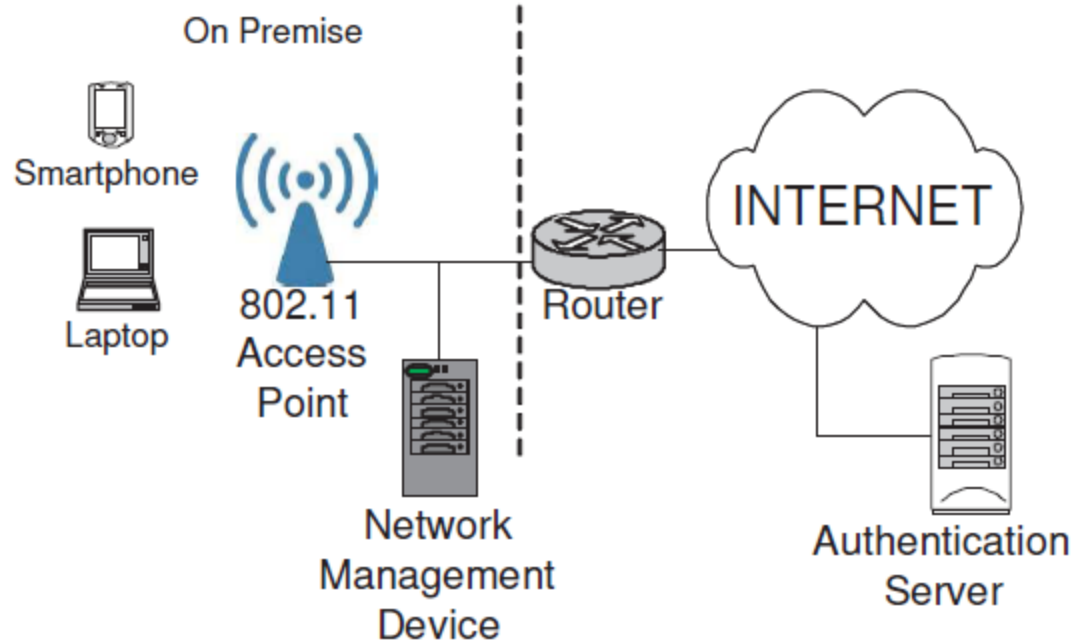
❑ **Realistic modeling of:**

- Session arrivals
- Connection duration distribution
- Simultaneously present customer distribution

**Capacity
Planning**



DATA COLLECTION



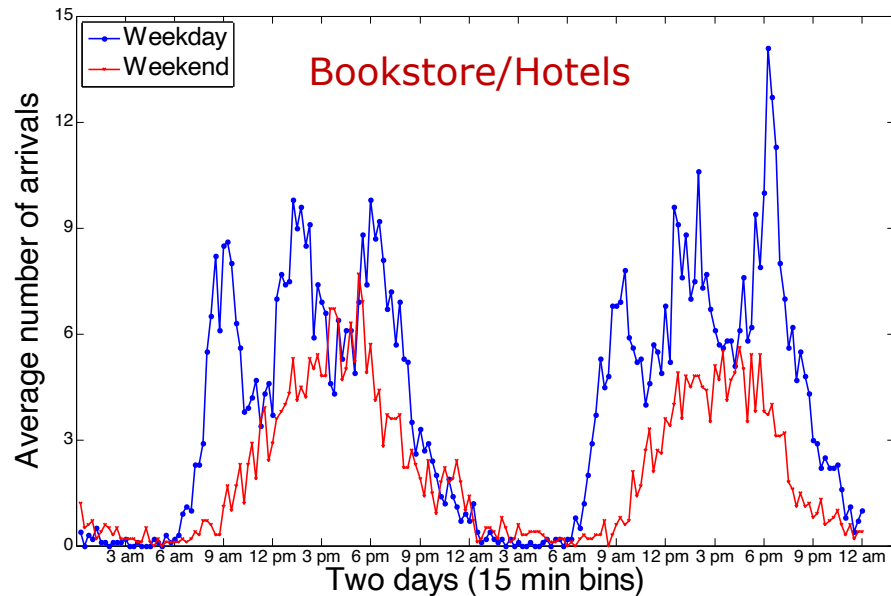
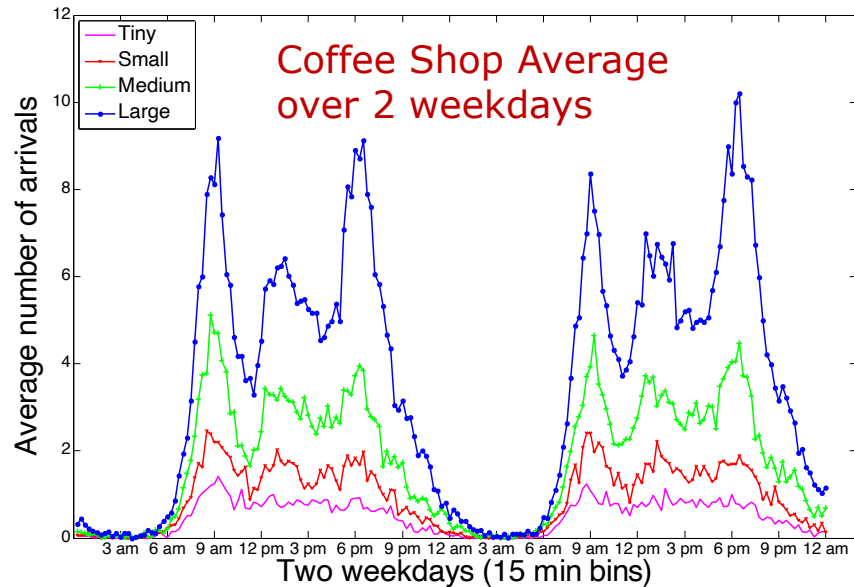
Mobile Internet Access using Wi-Fi hotspots



DATA STATISTICS

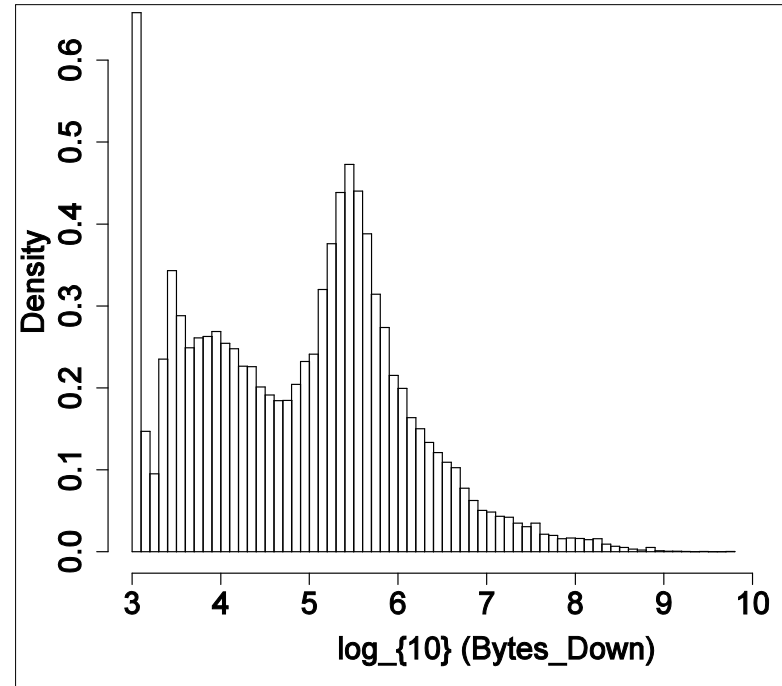
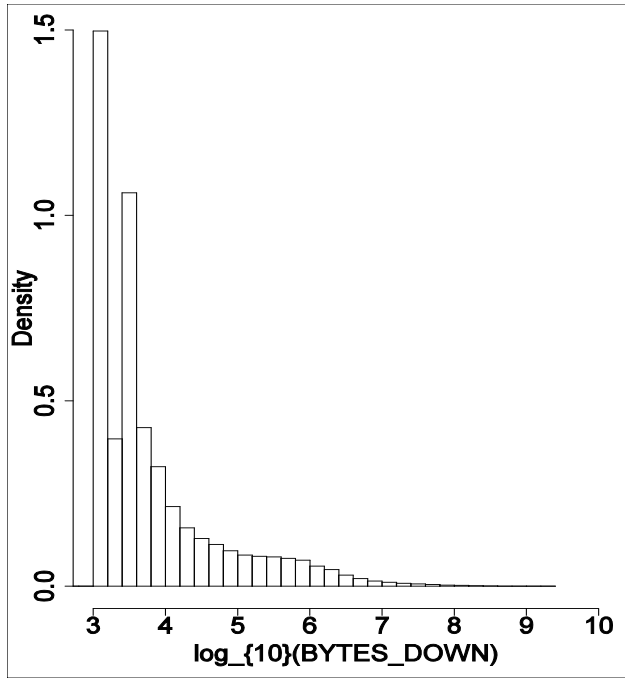
# of customers	234,742
# of devices	10
# of connections	1,322,541
# of cities	2 (NYC, SF)
# of Wi-Fi venues	362
# of zipcodes	87
Trace duration	4 weeks (3 weeks training, 1 week validation)

OVERVIEW: ARRIVALS



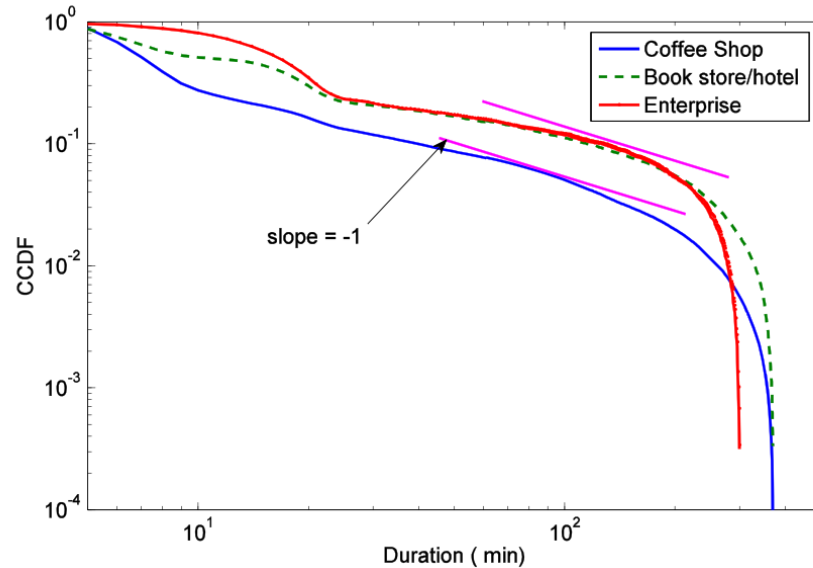
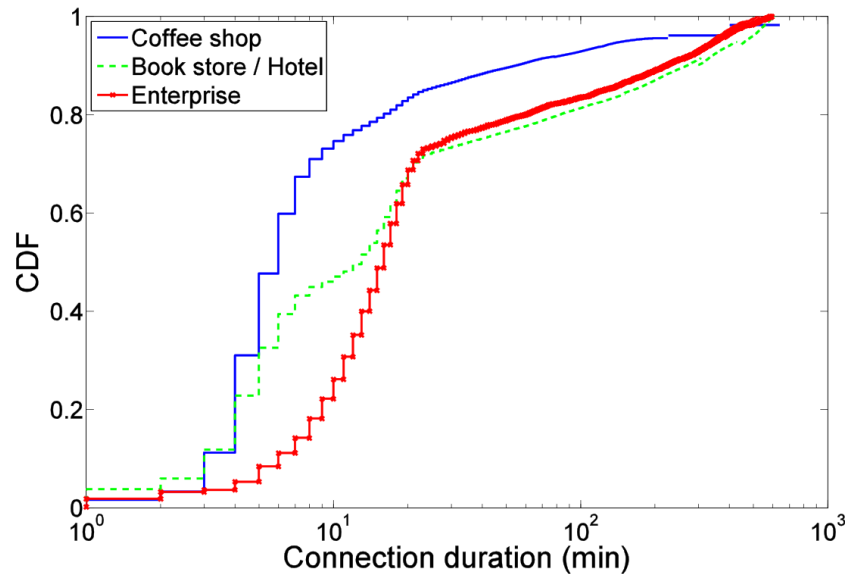
- ❑ Arrival rates vary drastically even within the same business type
- ❑ However, characteristic peaks in the means across all categories within the same business type
- ❑ Weekday and weekend patterns are significantly different => Need for different models

OVERVIEW: BYTE COUNTS



- ❑ Coffee shop customers typically download a few KB
- ❑ Enterprise customers typically download a few MB, and sometimes even a few GB
- ❑ Long tails

OVERVIEW: DURATIONS



- ❑ CDF of connection duration by business types
- ❑ Complimentary distribution function of connection durations (log-log scale) by business types => Long tails



ARRIVAL COUNT MODELING

□ Approach

- Markov Modulated Poisson Process is typically used to model arrival counts with “constant” periods of arrival rates.
- However, our data showed a “time-dependent” arrival rate.
- Polynomial curve fitting to the observed mean number of arrivals did not perform well since this method could not capture “within-day” pattern with a small number of terms.
- Standard Poisson regression also did not work well.
- Poisson regression with clustering was the preferred method. Noticed primarily two timescales in operation,
 - one of the order 1.5-3 hours (sudden rise and fall in the arrival rates)
 - and the other of the order 15-30 min (finer fluctuations)



ARRIVAL COUNT MODELING



❑ Non-stationary Poisson Process

- Time-dependent deterministic arrival rate
- Divide time into 3 hour bins **I: 8 bins per day**
- Divide each bin into 15 min slots **J: 12 slots per bin**

❑ Poisson Regression Model (GLM)

- Polynomial type dependence on bin and slot numbers

$$\log(\lambda(I, J)) = \alpha + \sum_{k=1}^3 \beta_k I^k + \sum_{k=1}^3 \gamma_k J^k + \delta(I * J)$$

where

first term: over-a-day mean behavior

sum terms: differential effects of specific cluster and slots within it

last term: interaction term – differential effect of slot J does not have to be the same across all clusters

ARRIVAL COUNT MODELING



❑ Clustering

- K-means clustering: Cluster time slots into groups such that within each group the average number of arrivals do not differ much
- Automatic 24 hour wrap-around in clustering

Cluster	Slots
0	1–28, 94–96
1	83–87
2	32–34, 38–40, 64–66
3	35–37, 72–75
4	48–58
5	41–47, 59–63, 67–71, 76–78
6	79–82
7	29–31, 88–93

❑ Clusters of 15 min time slots over a day

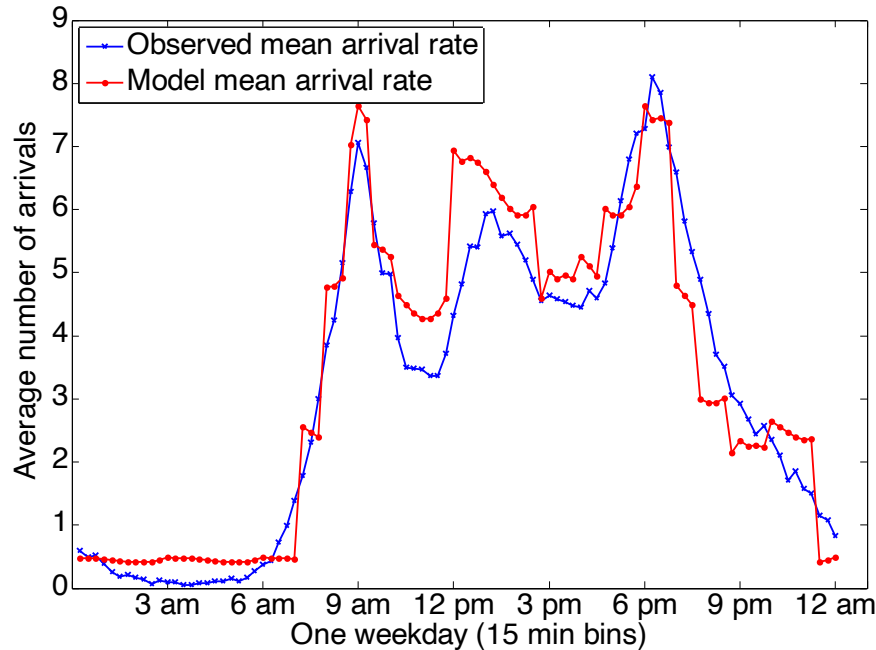
❑ Non-contiguous busy slots (35–37, 72–75) map to a common cluster

Heat map available???

RESULTS: ARRIVALS

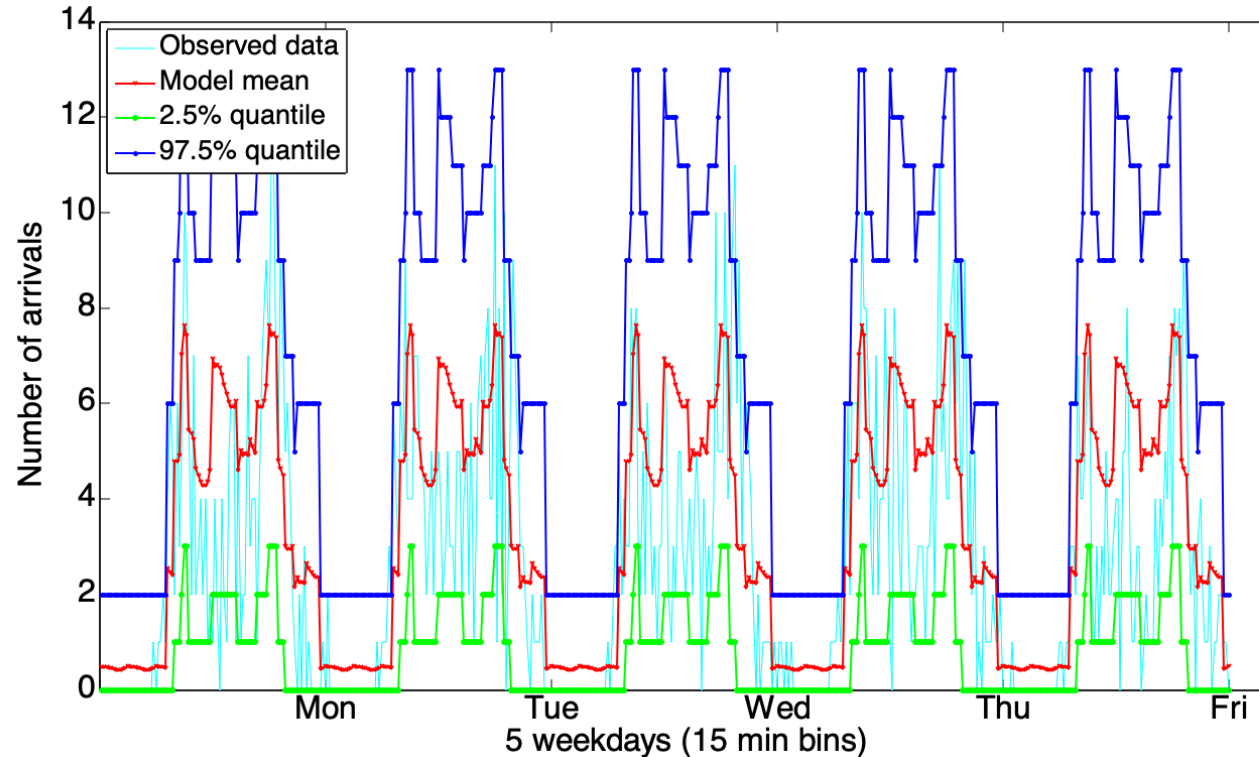


Show poorness of fit without clustering ???



❑ **Coffee shops:** Observed mean arrival rate plotted against the model mean arrival rate; these provide intra--day patterns for a cluster by averaging over its members

RESULTS: ARRIVALS



☐ **Coffee Shops:** Model mean arrival rate along with the 97.5% quantile and 2.5% quantile bands plotted against 5 days of validation data for an example coffee shop.

SESSION DURATION MODELING



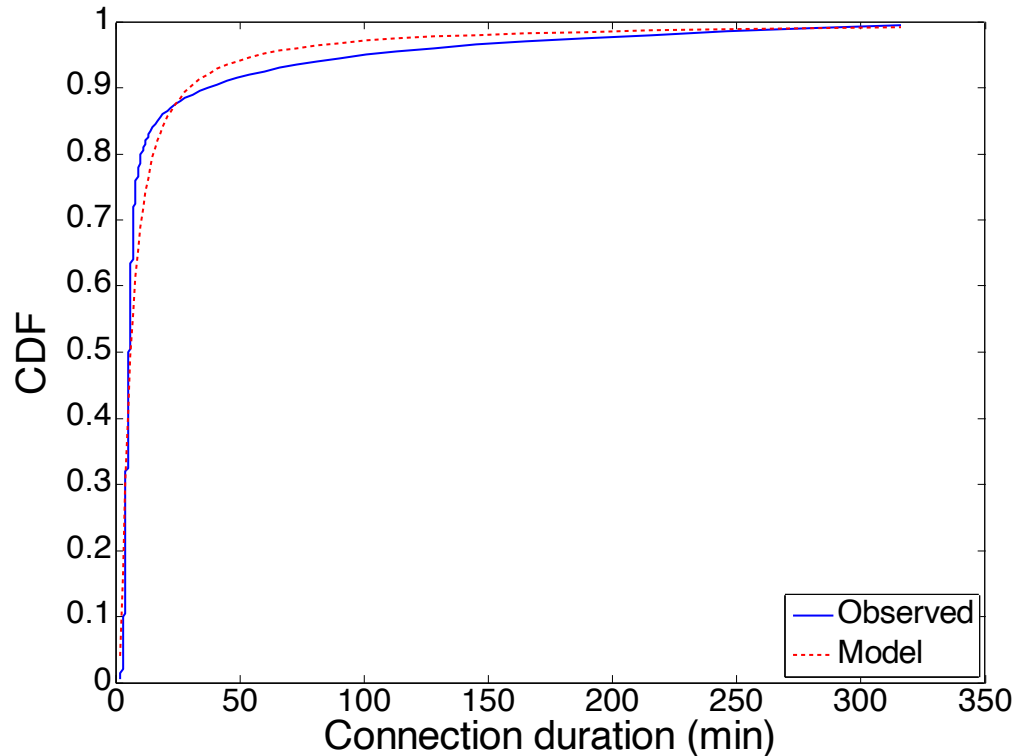
- Model the **logarithm of duration** (Y) as a **Phase-Type (PH) r.v.** (X)

$$Y \sim e^X$$

- **Properties of a PH-Type random variable**

- Distribution time to **absorption** in a suitably defined **Markov process**
- **Dense** in the class of all distributions, i.e., can approximate any distribution in $[0, \infty)$
- Exponentially decaying tail asymptotically going to 0 as $e^{-\eta x}$, where $-\eta$ is the real eigen-value of the rate transition matrix
- Captures not only the **tails** but also the **heads**, as opposed to **Pareto** and **Weibull**

RESULTS: DURATION



- ❑ Phase type distributions were fit using the EM algorithm
- ❑ A fit of order 5 was found to be adequate
- ❑ ANYTHING ELSE HERE??
- ❑ **Coffee Shops:** CDF plot of durations for coffee shops and data (truncated at 6 hours)

SIMULTANEOUS CONNECTIONS



- ❑ **Arrivals:** Non-homogeneous Poisson process (time-dependent arrival rates)
- ❑ **Durations (service times):** PH-type distribution
- ❑ **Simultaneously present customers:** $M_t/G/\infty$, queuing model

❑ **Theorem:**

The number of busy servers $Q(t)$ (i.e., number of simultaneously present customers) at time t follows a **Poisson distribution** with mean $m(t)$, where $H(.)$ is the service time distribution.

$$m(t) = \int_0^t \lambda(\tau)[1 - H(t - \tau)]d\tau$$

SIMULTANEOUS CONNECTIONS



- ❑ Novel proof based on **semi-regenerative type argument**
 - Does not require the system to be empty at some infinite past
 - Simple, transparent, and general

- ❑ Show that the **Probability Generating Function** $G(t)$ of $Q(t)$ is

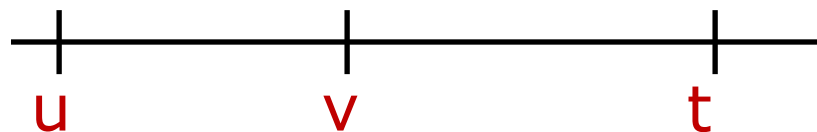
$$\begin{aligned} G_t(z, u) &= E[z^{Q(u)}] \\ &= e^{(z-1) \int_0^t \lambda(\tau) [1-H(t-\tau)] d\tau} \end{aligned}$$

SIMULTANEOUS CUSTOMERS



□ Proof idea:

- No arrivals in $(u, t]$
- First arrival occurs at some v in $(u, t]$
- $Q(u)$: No. of customers who arrive in $(u, t]$ and are still there at t
- $G(z, u)$: PGF of $Q(u, t)$
- Expected number of arrivals in $(0, t]$
- Then:



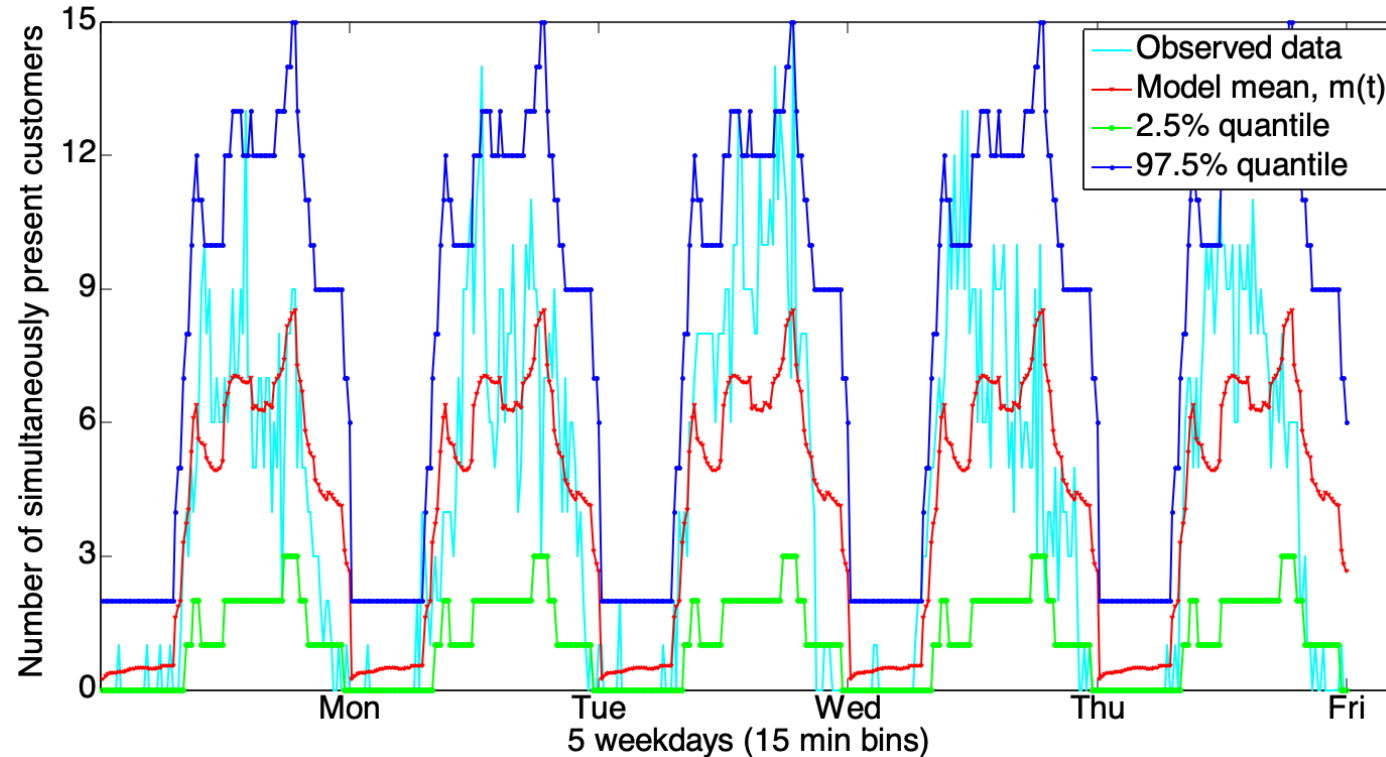
$$\Lambda(t) = \int_0^t \lambda(\tau) d\tau$$

$$G_t(z, u) = E[z^{Q(u)}]$$

Solve the integral equation

$$= e^{-[\Lambda(t)-\Lambda(u)]} + \int_u^t \lambda(v) e^{-[\Lambda(v)-\Lambda(u)]} \{H(t-v) + z[1-H(t-v)]\} G_t(z, v) dv$$

RESULTS: SIMULTANEOUS CUSTOMERS



❑ **Coffee Shops:** Expected number of simultaneously present customers along with the 97.5% quantile and 2.5% quantile bands plotted against 5 days of validation data for an example coffee shop.

CONCLUSIONS



- ❑ Examined salient differences w.r.t. arrival counts, temporal variations, connection durations, byte counts among venues of different business types
- ❑ Arrival count modeling using statistical clustering and non-stationary Poisson model under the GLM framework
- ❑ Use of phase-type random variables to model the logarithm of (long tailed) durations
- ❑ Estimating the number of simultaneously present customers using a queuing model
- ❑ New proof based on semi-regenerative argument for the number of busy servers in

$$M_t/G/\infty$$

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